

Day 1: Synthetic Division + Remainders

1. Use synthetic division:
- $(2x^3 + 5x^2 - 3x + 7) \div (x + 4)$

$$\begin{array}{r|rrrr} -4 & 2 & 5 & -3 & 7 \\ & & & & \end{array}$$

Answer: _____ + $\frac{\text{_____}}{x + 4}$

- 2.
- SAT:**
- When
- $p(x) = x^3 - 2x^2 + kx - 8$
- is divided by
- $(x - 2)$
- , the remainder is 0. What is
- k
- ?

A. -2 B. 2 C. 4 D. 6

Day 2: Box Method Division (Degree 4)

1. Use the box method:
- $(x^4 - x^3 + 2x^2 + 4x) \div (x - 3)$

x	x^4			
-3				

Answer: _____ + $\frac{\text{_____}}{x - 3}$

- 2.
- Spiral:**
- Simplify:
- $\frac{x^{n+2} \cdot x^{3-n}}{x^4} = \underline{\hspace{2cm}}$

Day 3: Sum & Difference of Cubes (with GCF)

1. Factor completely:
- $27x^3 - 125y^3 = \underline{\hspace{2cm}}$

2. Factor completely:
- $3x^4 - 24x = \underline{\hspace{2cm}}$

- 3.
- SAT:**
- Which is equivalent to
- $x^6 - y^6$
- ?

A. $(x^3 - y^3)(x^3 + y^3)$ B. $(x^2 - y^2)^3$ C. $(x - y)^6$ D. $(x^2 - y^2)(x^4 + y^4)$

- 4.
- SAT:**
- If
- $a^3 + b^3 = 35$
- and
- $a + b = 5$
- , what is
- $a^2 - ab + b^2$
- ?

A. 5 B. 7 C. 30 D. 175

Day 4: Factoring by Grouping (Degree 4)

1. Factor completely:
- $x^4 + x^3 - x - 1$
- (Hint:
- $x^3 - 1$
- is difference of cubes)

Answer: _____

- 2.
- SAT:**
- If
- $(x - 3)$
- is a factor of
- $x^3 + ax^2 - 7x + 6$
- , what is
- a
- ?

A. -4 B. -2 C. 2 D. 4

- 3.
- Spiral:**
- What is the value of
- $32^{-2/5}$
- ?

- A. $\frac{1}{4}$ B. $\frac{1}{8}$ C. $-\frac{4}{5}$ D. 4

Day 5: Mixed SAT/TSI Challenge

1. **SAT:** The polynomial $p(x)$ has factors $(x - 1)$, $(x + 2)$, and $(x - 3)$. Which could be $p(x)$?
A. $x^3 - 2x^2 - 5x + 6$ B. $x^3 + 2x^2 - 5x - 6$ C. $x^3 - 2x^2 + 5x - 6$ D. $x^3 + 2x^2 + 5x + 6$
2. Factor completely: $5x^3 + 10x^2 + 15x + 30 =$ _____
3. **SAT:** If $\sqrt[3]{by} = b^5$ and $a^{2x} \cdot a^3 = a^{15}$, what is $y - x$?
A. 6 B. 9 C. 12 D. 21
4. **TSI:** The expression $(x^{-5}y/y^3)^{-1}$ is equivalent to:
A. x^5y^2 B. $\frac{x^5}{y^2}$ C. $\frac{y^2}{x^5}$ D. x^5y^4